DIU-MKG: A Multimodal Knowledge Graph Dataset for Dynamic Instructional Understanding

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(Supplementary Material)

A Dataset Card and Release Overview

A.1 Dataset Overview

DIU-MKG is a multimodal knowledge graph dataset for dynamic instructional understanding in real classroom videos. Each sample is an episode-level graph aligned with a phase-bounded instructional segment. The dataset preserves concept progression, visual support, instructional emphasis, and learner-related cues in a unified structured form.

It supports research on structured instructional video understanding, especially where pedagogical organization cannot be reliably recovered from transcript input alone. It also provides reusable structured supervision for graph-grounded multimodal learning in educational scenarios.

A.2 Source and Scope

DIU-MKG is constructed from **463** K–12 teaching competition videos taught by frontline teachers, covering **10** subjects and about **95** hours of classroom recordings. The final release contains **1,852** episode graphs, **22,280** multimodal nodes, and **18,367** pedagogical relations. Table 1 summarizes the main dataset statistics.

The dataset adopts a lesson–phase–episode graph organization. Each lesson is segmented into four instructional phases: *introduction*, *exposition*, *interaction*, and *conclusion*. Each phase is instantiated as an independent graph unit.

A.3 Released Contents

The release includes graph annotations, aligned text fields, metadata, split files, and evaluation code. Depending on privacy constraints, raw source videos may be restricted. The released package is centered on graph-structured supervision and aligned multimodal context rather than unrestricted redistribution of classroom video data. The released components are summarized in Table 2. When raw video access is restricted, the released graph files and aligned multimodal context still allow reproducible study of structured pedagogical relations.

A.4 Intended Use and Limitations

DIU-MKG is intended for structured instructional understanding, including segment retrieval, pedagogical intent modeling, instructional question answering, pedagogical captioning, phase-aware retrieval, and graph-grounded multimodal learning. It is especially suitable for tasks that require explicit modeling of concept progression, pedagogical support, and instructional relations across classroom phases.

Table 1: Dataset card of DIU-MKG.

Field	Content
Dataset name	DIU-MKG
Task focus	Dynamic instructional understanding in real classroom videos
Source videos	463 K–12 teaching competition videos
Subjects	10 subjects
Total duration	95 hours
Graph units	1,852 episode graphs
Node count	22,280 multimodal nodes
Relation count	18,367 pedagogical relations
Graph organization	Lesson–phase–episode graph
Instructional phases	Introduction / Exposition / Interaction / Conclusion
Modal context	Transcript, visual facts, visual narrative, pedagogical goal, graph JSON
Intended use	Retrieval, intent recognition, instructional QA, pedagogical captioning, phase-aware retrieval, graph-grounded multimodal learning
Language	Chinese source classroom context; released graph labels in English

Because the dataset is derived from a specific real classroom context, conclusions should be interpreted with caution when generalized across regions, grade levels, or teaching settings. Access to raw video content may also be constrained by privacy policy.

B Graph JSON Schema

B.1 Graph-Level Structure

Each DIU-MKG sample is stored as a JSON object linking an instructional segment to its graph representation. At the graph level, a sample includes source metadata, aligned multimodal context, the graph object, and the graph-generator version. Figure 1 gives a compact view of the overall sample structure, while Table 3 lists the graph-level fields.

In internal files, the instructional-phase field is often stored as `teaching_stage`. This is an implementation detail and does not affect the lesson–phase–episode organization used in the paper.

B.2 Node Schema

Nodes represent the main semantic units in instruction. The current schema contains three node categories: concept, visual, and learner-state. In the current graph files, learner-state nodes are stored with the type value `Student_State`. Each node has a unique identifier and a concise label. When available, nodes also carry a timestamp aligned to the source segment. Table 4 summarizes the node schema.

Table 2: Released components and access scope in DIU-MKG.

Component	Description	Scope	Notes
Metadata	Subject, instructional phase, pedagogical goal, version fields	Released	Per-sample metadata
Transcript	Aligned teacher speech transcript	Released	Chinese source transcript
Visual facts	Structured visual description from key frames	Released	Used for graph construction
Visual narrative	Summarized visual evolution description	Released	Supports interpretation
Pedagogy graph JSON	Final nodes and edges	Released	Core annotation file
Split files	Retrieval / validation splits	Released	Task support
Evaluation code	Retrieval and ablation scripts	Released	Reproducibility support
Raw video / clips	Original classroom video content	Restricted / controlled	Subject to privacy and access policy

Table 3: Graph-level fields in DIU-MKG.

Field	Type	Description	Example
`video_name`	string	Source clip filename	`26_exposition.mp4`
`teaching_stage`	string	Instructional phase label of the segment; this field corresponds to the instructional phase in the lesson–phase–episode organization used in the paper	`exposition`
`subject`	string	Subject category	`biology`
`teaching_goal`	string	Segment-level pedagogical goal	Free text
`transcript`	string	Aligned transcript	Time-stamped utterances
`visual_facts`	string	Structured visual description	Free text
`visual_narrative`	string	Summarized visual evolution description	Free text
`pedagogy_graph`	object	Final graph object	JSON object
`graph_generator_version`	string	Version identifier	`pedagogy_graph_v7`

Table 4: Node schema in DIU-MKG.

Field	Applies to	Type	Description	Example
`id`	all nodes	string	Unique node identifier	`c5`, `v1`, `s1`
`label`	all nodes	string	Concise atomic label	`Extracellular Fluid`
`type`	all nodes	string	Node category	`Concept`, `Visual`, `Student_State`
`knowledge_type`	concept	string	Restricted knowledge-type label	`Fact`
`original_text`	visual	string	Raw or pre-correction visual description	Visual phrase / description
`ocr_corrected`	visual	boolean	Whether transcript-guided correction was applied	`true`
`cognitive_state`	learner-state	string	Learner-state category from a closed vocabulary	`Brainstorming`
`timestamp`	all node types (optional in some releases)	string	Segment-aligned time span	`[154.1 -> 322.94]`

B.3 Edge Schema

Edges are directed pedagogical relations connecting graph nodes. Each edge contains a source node, a target node, and a closed-set relation label. An optional `intent` field provides a short natural-language explanation. Table 5 lists the edge-level fields.

Table 5: Edge schema in DIU-MKG.

Field	Type	Description	Example
`source`	string	Source node identifier	`v1`
`target`	string	Target node identifier	`c5`
`relation`	string	Closed-set pedagogical relation label	`visualizes`
`intent`	string (optional)	Short explanation of the relation meaning	Free text

Table 6: Closed vocabularies used in DIU-MKG.

Dimension	Closed vocabulary
Node type	`Concept`, `Visual`, `Student_State`
Knowledge type	`Fact`, `Principle`, `Procedure`
Learner-state / `Student_State` type	`Brainstorming`, `Answering`, `Questioning`, `Confused`, `Engaged`
Teacher–visual relations	`visualizes`, `scaffolds`, `summarizes`, `emphasizes`
Teacher–student relations	`prompts_thinking`, `praises_response`, `guides_discussion`, `addresses_misconception`
Knowledge-evolution relations	`evolves_from`, `builds_upon`, `includes`

B.4 Closed Label Spaces

To keep the schema stable, DIU-MKG uses closed vocabularies for node categories, knowledge types, learner-state attributes, and pedagogical relations. These vocabularies are summarized in Table 6.

B.5 Sample Structure Illustration

Figure 1 illustrates the structure of a DIU-MKG sample, including source metadata, aligned multimodal context, and the final pedagogical graph.

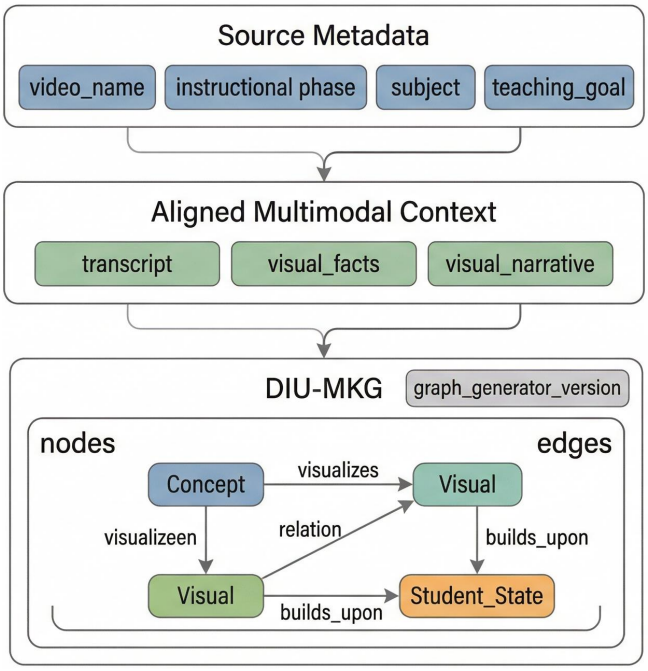

Figure 1: Structure of a DIU-MKG sample. Each sample stores source metadata, aligned multimodal context, and the final pedagogical graph with nodes and directed relations.

C Prompting Protocol for Graph Construction

C.1 Input Context

Graph construction in DIU-MKG is conditioned on five input fields: **instructional phase**, **pedagogical goal**, **visual facts**, **visual narrative**, and **aligned transcript**. Together they provide pedagogical context, visual evidence, and linguistic grounding.

The transcript serves as the main semantic anchor, while visual facts and visual narrative describe board writing, slides, diagrams, and classroom actions.

C.2 Prompt Design Objectives

The prompting protocol follows five basic constraints: (1) output in English; (2) keep node labels short and atomic; (3) preserve graph density rather than collapsing content into a few coarse nodes; (4) repair noisy visual parses using transcript grounding; and (5) return valid JSON that matches the released schema. Table 7 summarizes these constraints.

C.3 Cross-Modal Refinement Rule

A central part of the prompting protocol is transcript-guided cross-modal refinement. When the visual parse conflicts with the transcript or broader visual narrative, the model is instructed to correct the visual interpretation rather than preserve the raw OCR description. Corrected visual nodes are marked with `ocr_corrected = true`. In the current release, this transcript-guided refinement affects 31.78% of visual nodes, consistent with the statistic reported in the main paper.

C.4 Atomic Node Granularity

Node labels are kept short, and complex instructional content is decomposed into smaller graph units linked by explicit relations. Table 8 gives representative examples.

C.5 Output Validation and Post-Processing

Model outputs are subjected to JSON parsing and structural validation before being retained in the released dataset. Malformed outputs or responses that fail schema-level checks are discarded and regenerated. Valid graph files are written to a mirrored output directory rather than overwriting the raw annotations.

Table 7: Prompt constraints and their purposes in DIU-MKG graph construction.

Constraint	Purpose	Example
English-only output	Unify released graph labels and avoid multilingual inconsistency	Translate Chinese transcript content into English graph labels
Atomic node labels	Prevent sentence-like nodes and improve compositional reuse	`Extracellular Fluid` instead of a long sentence
Dense graph requirement	Preserve instructional structure	Split concept, visual support, and learner activity into separate nodes
Transcript-guided correction	Repair noisy visual parse using a stable anchor	Correct a noisy board fragment into `Mass Spectrometer`
Strict JSON output	Ensure machine-readable release files	Nodes / edges returned as a JSON object
Closed label sets	Improve schema consistency	Restrict `knowledge_type` to Fact / Principle / Procedure

Table 8: Examples of atomic node decomposition in DIU-MKG.

Raw expression	Undesirable node label	Preferred atomic decomposition
Teacher explains the exchange between intracellular fluid and extracellular fluid	`Teacher explains intracellular and extracellular fluid exchange`	`Intracellular Fluid` + `Extracellular Fluid` + `Fluid Exchange Process`
Students brainstorm measures to prevent land pollution	`Students brainstorming how to prevent land pollution`	`Preventing Land Pollution` + learner-state node `Brainstorming`
Mass spectrometer separates particles through different trajectories	`Mass spectrometer separates particles by trajectory difference`	`Mass Spectrometer` + `Particle Separation` + `Different Trajectories`

C.6 Full Prompt Template

Figure 2 shows the full prompt template used in the graph construction pipeline. We keep the implementation wording and output constraints to reflect the actual generation setup as closely as possible.

D Cross-Modal Refinement Example

D.1 Representative Case: OCR Ambiguity in a Physics Exposition Segment

A representative physics exposition segment illustrates transcript-guided refinement. In the raw visual parse, some board-level strings are semantically unstable, including *"different tracks → electrons separate"* and *"different wheels → electric field"*. These strings are visually plausible OCR fragments, but they are not scientifically coherent in the local lesson context.

The aligned transcript makes the local teaching intent clear: the teacher explains that charged particles can be separated because their trajectories differ under electric and magnetic fields, and then introduces the operating principle of the mass spectrometer. Figure 3 visualizes this refinement process, and Table 9 summarizes the corresponding corrections.

D.2 Summary of Refinement Patterns

Across the dataset, transcript-guided refinement mainly addresses four recurring error types: OCR noise, semantically incomplete handwritten fragments, formula corruption, and discipline-specific term ambiguity. Table 10 summarizes these patterns.

E Complete Annotation Examples

E.1 Example 1: Mathematics Exposition Episode

This example corresponds to a mathematics lesson in the **exposition** phase. The segment introduces the concept of sequences, explains that sequence terms are arranged in a fixed order, and distinguishes between finite and infinite sequences.

The teacher uses height records and daily lunar visibility as concrete examples, then moves to the formal definition of a sequence and the distinction between finite and infinite sequences. Visual support is provided through blackboard definitions and screen-based examples. Figure 4 shows the resulting annotation.

E.2 Example 2: Physics Interaction Episode

This example corresponds to a physics lesson in the **interaction** phase. The segment formalizes the concept of momentum, introduces the formula $p = mv$, and guides students to identify its direction, unit, vector nature, and relative character through discussion. Figure 5 shows the resulting graph annotation.

E.3 Annotation Rationale

These examples illustrate the main annotation principles in DIU-MKG: concepts are decomposed into atomic nodes, visual resources

Full prompt template used in the DIU-MKG graph construction pipeline

Template

```
You are an expert Data Annotator for the "PedagogyGraph" dataset targeting the ACM Multimedia Dataset Track.
Your task is to construct a 4D Cognitive-Dynamic Knowledge Graph from a Classroom Video.

[Input Context]
- Teaching Phase: {stage}
- Pedagogical Goal: {teaching_goal}
- Visual Facts: {facts}
- Visual Narrative: {clean_narrative}
- Transcript (Chinese):
{clean_transcript}
————————

## CRITICAL RULE 1: STRICTLY ENGLISH ONLY
Although the Transcript is in Chinese, the ENTIRE OUTPUT JSON (including all label, original_text, intent, and any other fields) MUST BE TRANSLATED INTO
PROFESSIONAL ACADEMIC ENGLISH. Do not output any Chinese characters under any circumstances.

## CRITICAL RULE 2: ATOMIC NODES AND DENSER GRAPH
- Node labels MUST be extremely concise and atomic (1 to 5 words max).
- Do NOT write long sentences in nodes.
- If a concept or interaction is complex, BREAK IT DOWN into multiple smaller, interconnected nodes to create a larger, denser graph.
* Bad Label: "Students brainstorming solutions for papaya virus" (Too long)
* Good Label: "Brainstorming Solutions" (Atomic) -> connected to -> "Papaya Virus" (Atomic)
————————

## CRITICAL INSTRUCTION: CROSS-MODAL REFINEMENT
The 'Visual Facts' (OCR) may contain hallucinations. Use the 'Transcript' as the ground truth semantic anchor:
- If OCR text conflicts with the Transcript/Narrative, correct it and translate it to English.
- Set "ocr_corrected": true for any Visual node where you fixed such errors.
————————

## TASK: EXTRACT A COGNITIVE-DYNAMIC GRAPH

## TASK: EXTRACT A HIGH-DENSITY COGNITIVE-DYNAMIC GRAPH
(EXTREMELY IMPORTANT: DO NOT SUMMARIZE. YOU MUST BE EXHAUSTIVE.)

### 1. Nodes (Entities and Actors)
- Type 'Concept': Abstract knowledge.
- label: Short English academic term (1-5 words max).
- CRITICAL: You MUST extract EVERY specific concept mentioned in the Transcript. For example, if the teacher lists "anthrax, smallpox, cholera", you MUST
create 3 separate concept nodes. Do NOT group them into a single "Pathogens" node. Aim for at least 10-15 concept nodes for a 2-minute video.
- knowledge_type: Choose STRICTLY from ["Fact", "Principle", "Procedure"].
- timestamp: Time range.
- Type 'Visual': Concrete screen/board element.
- label: Short English academic description. CRITICAL: Translate vague terms like "New Slide Columns" into "Weapon Comparison Slide" based on what the teacher
is saying.
- original_text: Translated English description.
- ocr_corrected: boolean.
- Type 'Student_State': Students' cognitive status.
- label: Concise description (e.g., "Group Brainstorming").
- cognitive_state: Choose from ["Brainstorming", "Answering", "Questioning", "Confused", "Engaged"].

### 2. Edges (Pedagogical Dynamics)
Edges MUST flow logically (e.g., Concept -> Visual, or Concept -> Concept). Create a densely connected graph.
- relation: Choose STRICTLY from:
- Teacher-Visual: "visualizes", "scaffolds", "summarizes", "emphasizes"
- Teacher-Student: "prompts_thinking", "praises_response", "guides_discussion", "addresses_misconception"
- Knowledge-Evolution: "evolves_from", "builds_upon", "includes" (use this to connect parent concepts to specific examples).
- intent: A clear English explanation.

## OUTPUT: STRICT JSON ONLY
{
"nodes": [
{
"id": "c1", "label": "Transgenic Papaya", "type": "Concept",
"knowledge_type": "Fact", "timestamp": "[220.72 -> 251.88]"
}
],
"edges": [
{
"source": "c1", "target": "c2", "relation": "builds_upon",
"intent": "Links the specific crop to the underlying biological principle."
}
]
}
```

Figure 2: Full prompt template used in the DIU-MKG graph construction pipeline. Placeholder variables such as `{stage}`, `{facts}`, and `{clean_transcript}` are filled dynamically for each instructional segment.

are modeled explicitly when they play a pedagogical role, learner-state nodes are included when student response is observable, and relations are used to encode instructional dependencies rather than flat co-occurrence. In Figures 4 and 5, concept, visual, and learner-state nodes are color-coded for readability. The rationale for representative components is summarized in Table 11.

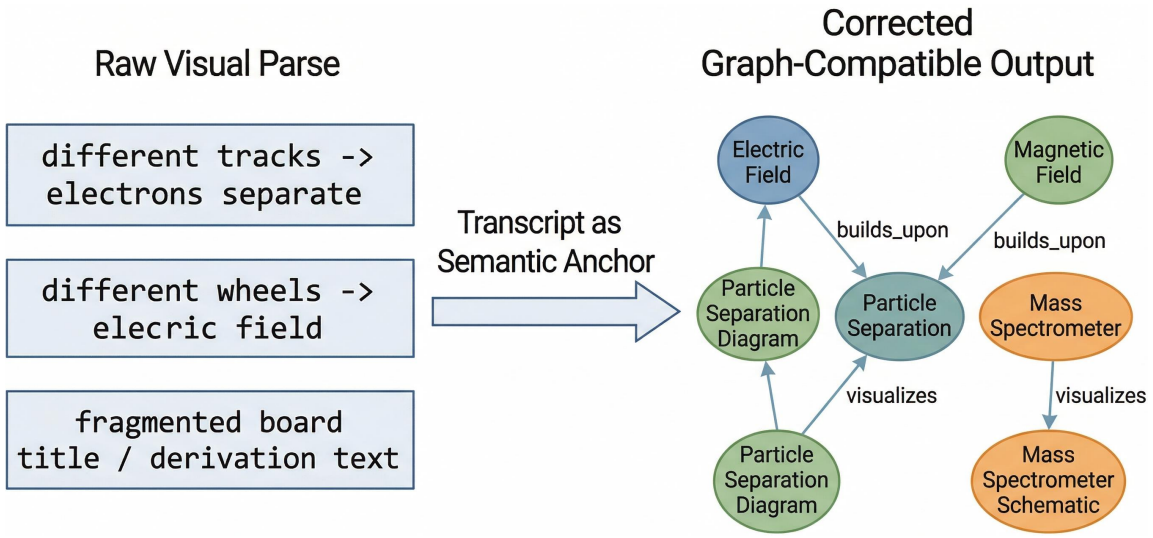

Figure 3: Transcript-guided cross-modal refinement in a physics exposition segment. Noisy visual parses are corrected using the transcript as a semantic anchor, yielding graph-compatible concepts and visual nodes.

Table 9: Representative transcript-guided correction case from a physics exposition segment.

Raw visual parse	Transcript cue	Corrected node / concept	Why correction is needed
"Different tracks → electrons separate"	The teacher explains that different particles can be separated because their trajectories differ.	**Particle Separation** / **Different Trajectories**	The OCR fragment is incomplete and incorrectly introduces "electrons."
"Different wheels → electric field"	The teacher explains that charged particles experience force in an *electric field* or *magnetic field*.	**Electric Field** / **Magnetic Field**	The visual parse contains a nonsensical lexical fragment.
Board title and derivation fragments	The teacher explicitly introduces the *mass spectrometer* and derives the relation between m, v, q, B, and r.	**Mass Spectrometer Schematic** / $r = \frac{mv}{qB}$	The transcript anchors the scientific topic and filters out noisy board fragments.

Table 10: Typical transcript-guided refinement patterns in DIU-MKG.

Error source	Raw visual issue	Transcript cue	Correction effect
Board OCR noise	Garbled lexical fragment	Teacher explicitly names concept	Recover stable concept node
Handwritten ambiguity	Incomplete phrase / unclear symbol	Local sentence-level explanation	Restore pedagogically meaningful label
Formula corruption	Missing / malformed variable	Formula verbally derived in transcript	Repair scientific expression
Term ambiguity	Wrong domain term or entity	Transcript names correct term	Reduce node-level semantic mismatch

F Expert Validation Criteria and Results

F.1 Validation Setup

We conduct expert validation on **300** sampled episode graphs reviewed by **three** PhD students with educational backgrounds. Each sampled graph is examined together with its aligned instructional segment, including transcript and visual context.

F.2 Review Checklist

The review focuses on node validity, relation validity, and overall graph coherence. Table 12 shows the checklist used in the review.

F.3 Validation Metrics

We summarize the review outcomes with overall acceptance rate and expert consistency. The final statistics are shown in Table 13.

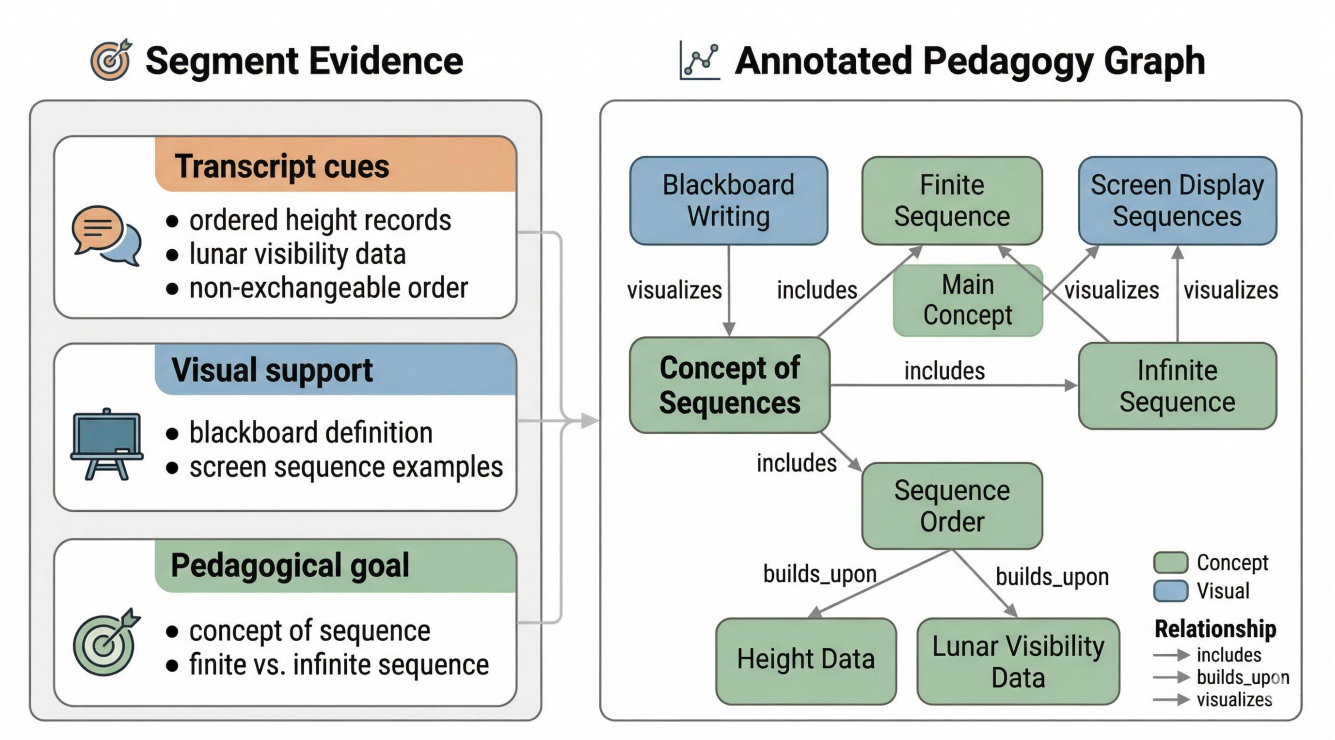

Figure 4: Complete annotation example from a mathematics exposition segment. The graph links real-world sequence examples, formal definitions, and category distinctions through explicit pedagogical relations.

Table 11: Annotation rationale for the representative mathematics and physics examples.

Graph component	Evidence source	Why included	Pedagogical role
Sequence Order	Transcript + definition writing	The teacher repeatedly stresses non-exchangeable order	Captures the defining property of sequences
Height Data / Lunar Visibility Data	Transcript examples	Concrete real-world data used to motivate abstraction	Supports example-based concept formation
Momentum Formula	Transcript + chalkboard	The teacher explicitly formalizes $p = mv$	Converts observation into formal concept
Vector Quantity	Transcript questioning	Introduced through teacher-led conceptual refinement	Captures property-level elaboration
Student Answering	Classroom dialogue	Student responses are explicitly solicited and evaluated	Represents interactive knowledge construction
Visual nodes in both examples	Blackboard / screen descriptions	Visual resources anchor definitions and examples	Support explanation, emphasis, and reinforcement

F.4 Interpretation of Results

The sampled graphs are generally well aligned with the source videos and show good semantic validity under real classroom conditions. The remaining disagreements are concentrated in segments with noisy visual evidence, ambiguous handwritten content, or fine-grained pedagogical distinctions.

G Error Taxonomy and Representative Failures

G.1 Error Taxonomy Overview

The remaining errors in DIU-MKG can be grouped into three categories: (1) global semantic misalignment between transcript-grounded lesson content and the visual channel, (2) cross-topic contamination caused by visually retained but off-topic slide or board content, and (3) relation-level semantic drift, where node

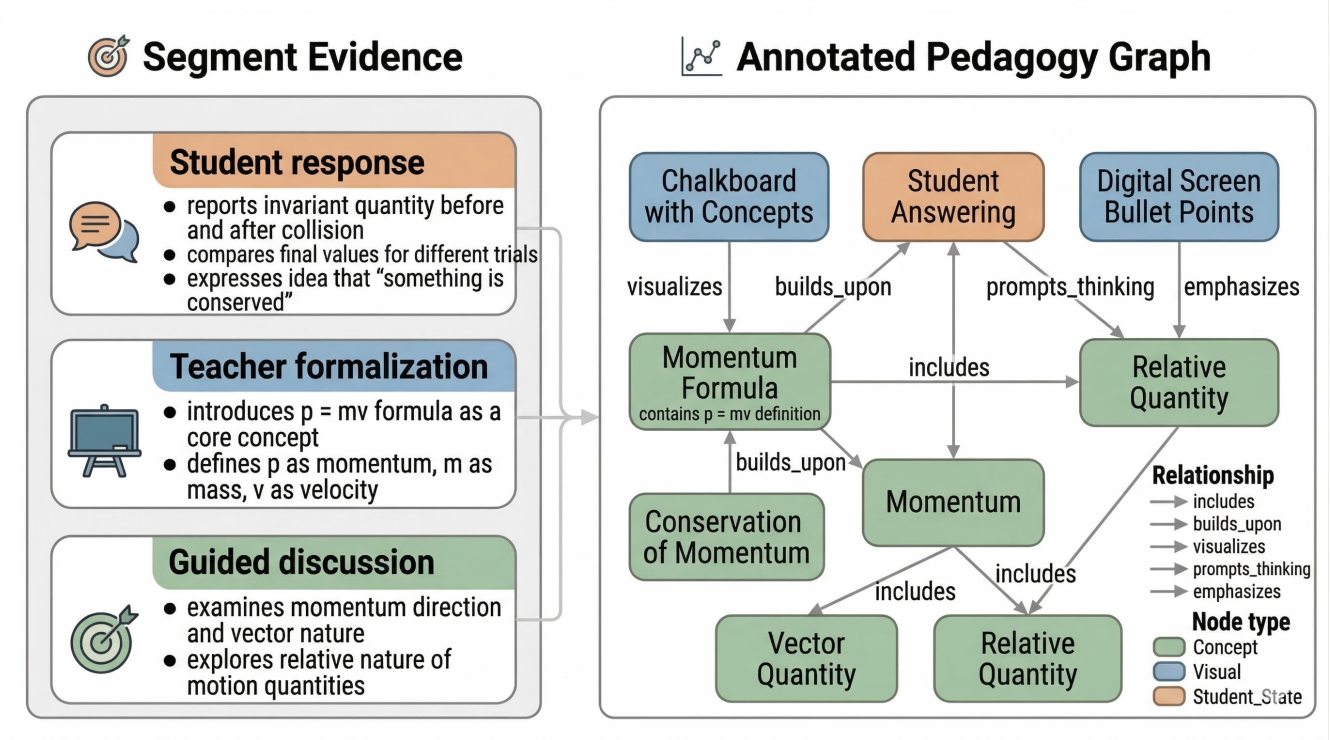

Figure 5: Complete annotation example from a physics interaction segment. The graph highlights how student responses, board writing, and formula formalization jointly support concept construction around momentum.

Table 12: Expert review checklist used in DIU-MKG validation.

Dimension	Review question	Acceptance criterion
Node validity	Are the extracted concepts, visual nodes, and learner-state cues semantically consistent with the source segment?	Major nodes are semantically valid and appropriately typed
Relation validity	Are the relation labels and directions compatible with the local classroom context?	Relations are pedagogically plausible and interpretable
Graph coherence	Does the graph preserve the major pedagogical structure of the segment?	Overall graph is acceptable as a structured representation of the segment

Table 13: Expert validation summary for DIU-MKG.

Metric	Value
Sampled episode graphs	300
Reviewers	3
Overall acceptance rate	93.3%
Expert consistency	92.7%

recovery is partly correct but the final pedagogical relations remain unstable. These categories are summarized in Table 14.

G.2 Failure Type I: Global Semantic Misalignment

A representative example comes from an English interaction segment. The teaching goal concerns culture-related vocabulary and image-based cultural description, while the visual context instead introduces unrelated concepts such as *Choice Voting System*, *Referendum*, and *Bias for*. The generated graph therefore follows the wrong visual frame rather than the local instructional content.

G.3 Failure Type II: Cross-Topic Contamination

A representative example comes from a biology conclusion segment whose transcript discusses applications of genetic engineering, while the visual channel preserves unrelated concepts such as *natural selection*, *adaptive evolution*, and *trait selection*. The resulting graph mixes the intended summary with off-topic biological evolution content.

G.4 Failure Type III: Relation-Level Semantic Drift

A representative example comes from a physics exposition segment about charged particles, electric and magnetic fields, and the mass spectrometer. Transcript grounding helps recover correct high-level

Table 14: Error taxonomy for representative DIU-MKG failure cases.

Error type	Description	Typical cause	Affected component
Global semantic misalignment	The graph follows an irrelevant visual frame instead of the actual lesson semantics in the transcript	Severe visual-text mismatch or wrong visual summary context	Concept nodes, visual nodes, whole-graph topic
Cross-topic contamination	Part of the target topic is recovered, but unrelated slide or board content is also incorporated	Visual context contains off-topic or retained previous-slide information	Concept set, graph purity, summary semantics
Relation-level semantic drift	Core concepts are mostly correct, but edge labels or directions are scientifically or pedagogically unstable	Weak relation grounding under noisy multimodal evidence	Edge labels, edge directions, graph interpretability

nodes, but some final relations remain semantically unstable, for example by incorrectly linking fields to mass or charge properties.

Table 15: Automatic and expert-involved stages in the DIU-MKG workflow.

Stage	Mode	Purpose
Transcript / visual loading	Automatic	Read aligned multimodal context
Sentence-aware truncation	Automatic	Reduce prompt overflow while preserving semantics
Prompt construction	Automatic	Assemble graph-generation input
LLM graph generation	Automatic	Produce graph JSON from multimodal context
JSON parsing / validation	Automatic	Retain only structurally valid outputs
Mirrored output writing	Automatic	Preserve raw files and version generated graphs
Sampled review	Expert-involved	Check graph semantic validity
Release filtering	Hybrid	Decide what is included in the released package

Table 16: Task specification summary for the evaluation settings used in DIU-MKG.

Task	Input	Output	Metric
Retrieval	Query + 5 candidate segments	Correct target segment	Accuracy (%), mean±std
Relation ablation for intent recognition	Transcript / graph variants	Predicted pedagogical intent	Accuracy (%), mean±std

G.5 Discussion of Error Boundaries

These failures mainly reflect the difficulty of extracting stable pedagogical structure from noisy real classroom signals. In particular, they show that transcript and visual context may diverge globally, conclusion segments are vulnerable to retained non-core visual content, and relation-level semantics remain harder to stabilize than node-level semantics.

Illustrative Retrieval Instance

Query
Which segment explains that momentum is a vector quantity whose direction follows velocity?

Candidate Segments
- Segment A (target)
- Segment B
- Segment C
- Segment D
- Segment E

Correct Answer
Segment A

Figure 6: Illustrative retrieval query and candidate setup.

H Human-in-the-Loop Construction Workflow

H.1 Pipeline Overview

DIU-MKG is constructed through a semi-automatic human-in-the-loop workflow that combines automatic multimodal parsing with expert quality control. The overall process includes source annotation loading, transcript and visual-context preprocessing, LLM-based graph generation, JSON-level structural validation, version-controlled output storage, global statistics collection, expert review on sampled outputs, and release filtering.

H.2 Automatic Stages

The automatic stages include: loading source annotations; sentence-aware truncation of long transcript and visual narrative fields; prompt construction using instructional phase, pedagogical goal, transcript, and visual context; GPT-4o-based graph generation; JSON parsing and structural validation; and mirrored output writing.

H.3 Expert-Involved Stages

Expert involvement is introduced during quality assurance rather than full manual graph construction. Reviewers examine sampled graph outputs together with aligned source evidence to assess node validity, relation plausibility, and overall pedagogical coherence. The main stages are summarized in Table 15.

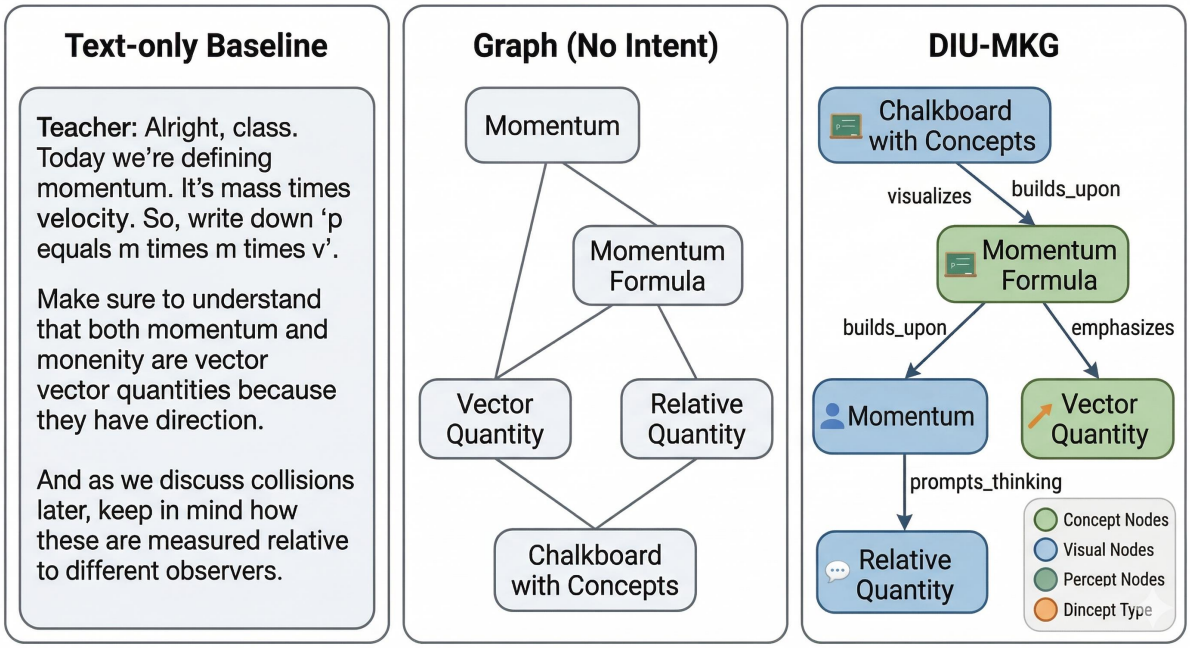

Figure 7: Comparison of the three evaluation inputs used in relation ablation: Text-only Baseline, Graph (No Intent), and DIU-MKG. The full DIU-MKG setting preserves explicit pedagogical relation semantics.

H.4 Scalability Considerations

Compared with fully manual graph labeling, the semi-automatic workflow substantially reduces annotation burden while keeping the construction process scalable to real classroom video collections. Sampled expert review provides an empirical basis for reporting dataset reliability.

I Task Specifications for Retrieval and Relation Ablation

I.1 Retrieval Task Definition

The zero-shot instructional segment retrieval task evaluates whether DIU-MKG supports segment-level pedagogical semantic matching. Given a query constructed from the target segment, the model must identify the correct instructional segment from a candidate set that includes the target and several distractors. Table 16 summarizes the evaluation settings.

I.2 Retrieval Query and Candidate Construction

In the current implementation, retrieval queries are generated from the core concepts and pedagogical relations of the target segment. Each candidate set contains **5** instructional segments, including the target and **4** distractors. The task is repeated over **200** generated queries, and results are averaged over **10** runs.

Figure 6 gives a simplified retrieval example.

For the DIU-MKG setting, the graph is linearized into structured textual input that preserves node content and relation semantics. In the text-only baseline, only the transcript is provided.

I.3 Ablation Task Definition

The ablation setting isolates the contribution of pedagogical relation semantics. Three settings are compared: (1) **Text-only Baseline**, which uses only transcript input; (2) **Graph (No Intent)**, which preserves graph nodes and connectivity while removing relation semantics; and (3) **DIU-MKG**, which preserves the full graph structure and relation semantics.

I.4 Example Inputs for the Three Settings

Figure 7 compares the three input settings used in the ablation study. The contrast makes clear what information is removed or preserved at each stage, especially the role of explicit pedagogical relation semantics in the full DIU-MKG setting.

I.5 Additional Notes on Future Tasks

Beyond the current settings, DIU-MKG can also support instructional question answering, pedagogical captioning, phase-aware retrieval, temporal instructional grounding, and graph-grounded multimodal generation. Representative downstream extensions are listed in Table 17.

Table 17: Potential downstream tasks supported by DIU-MKG.

Task	Output	Why DIU-MKG supports it
Instructional QA	Answer span / rationale	Graph preserves concepts, visual support, and pedagogical relations
Pedagogical captioning	Structured caption	Graph offers explicit node-relation supervision
Phase-aware retrieval	Segment ranking	Episode graphs are phase-aligned by design
Temporal grounding	Time span / segment	Graphs preserve alignment with phase-bounded clips
Graph-grounded generation	Structured explanation	Relations provide reusable instructional structure